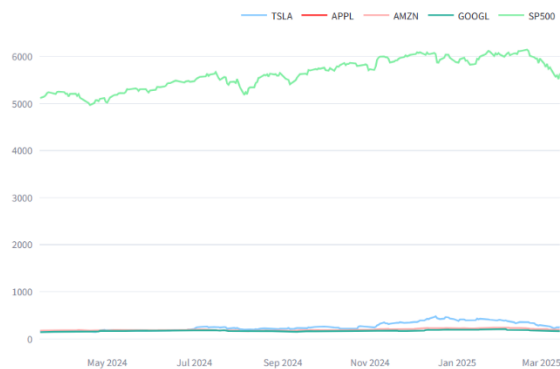


Capital Asset Pricing Model

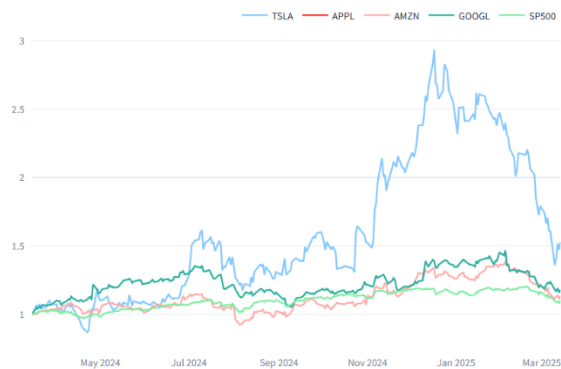
The CAPM Financial Analysis project is a Python-based implementation of the Capital Asset Pricing Model ([CAPM](#)). This model is widely used in finance to estimate the expected return on an investment by taking into account the risk-free rate, the market risk premium, and the asset's *beta*. The project provides functionalities to calculate the expected returns for a given set of securities based on the CAPM formula.

The application uses libraries such as Pandas, NumPy, Streamlit and Plotly, to gather stock data from Yahoo Finance and perform calculations to determine expected returns.

Price of all stocks



Price of noramlized stocks



Capital Asset Pricing Model

Choose 4 stocks

TSLA x APPL x AMZN x GOOGL x

Number of years:

1

Dataframe Head

	Date	TSLA	APPL	AMZN	GOOGL	SP500
0	2024-03-15 00:00:00	163.57	None	174.42	140.5082	5117.09
1	2024-03-18 00:00:00	173.8	None	174.48	146.9773	5149.42
2	2024-03-19 00:00:00	171.32	None	175.9	146.3304	5178.51
3	2024-03-20 00:00:00	175.66	None	178.15	148.0323	5224.62
4	2024-03-21 00:00:00	172.82	None	178.15	146.8977	5241.53

Dataframe Tail

	Date	TSLA	APPL	AMZN	GOOGL	SP500
245	2025-03-10 00:00:00	222.15	None	194.54	165.87	5614.56
246	2025-03-11 00:00:00	230.58	None	196.59	164.04	5572.07
247	2025-03-12 00:00:00	248.09	None	198.89	167.11	5599.3
248	2025-03-13 00:00:00	240.68	None	193.89	162.76	5521.52
249	2025-03-14 00:00:00	249.98	None	197.95	165.49	5638.94

Calculated Beta Value

	Stock	Beta Value
0	TSLA	2.73
1	APPL	nan
2	AMZN	1.43
3	GOOGL	1.22

Calculated return using CAPM

	Stock	Return Value
0	TSLA	29.3
1	APPL	nan
2	AMZN	15.33
3	GOOGL	13.1

Usage

The CAPM Financial Analysis application allows you to select a list of stocks and specify the number of years for investment. It then calculates the expected returns for each stock based on the CAPM formula and displays the results. Follow these steps:

- Choose stocks by ticker:
 - Use the multi-select input to choose one or more stocks by their tickers.
 - Available stock tickers are provided in the dropdown menu.
- Specify the number of years of investment:
 - Use the number input to enter the number of years you plan to invest.
- View the results:
 - The application retrieves historical stock price data for the selected stocks and the [S&P 500](#) index.
 - It normalizes the stock prices based on the initial prices and displays the normalized prices in an interactive plot.
 - It calculates the beta values and alpha values for each stock based on the daily returns.
 - The calculated beta values and alpha values are displayed in a table.
 - The application calculates the expected returns using the CAPM formula and displays them in a table.
- Interpret the results:
 - The normalized prices plot helps visualize the performance of selected stocks relative to the initial prices.

- The calculated beta values indicate the riskiness of each stock relative to the market (S&P 500).
- The expected returns table shows the estimated returns for each stock based on the CAPM formula.

Conclusion

The CAPM Financial Analysis application provides an interactive interface for calculating expected returns based on CAPM. By selecting stocks and specifying the investment period, users can estimate the returns and analyze the riskiness of their investment choices.